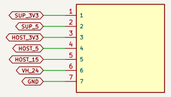
DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / NATIVE INDEPENDENT SUPERVISION

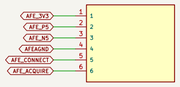
18 guarded stimulation conductors, 17 measured RHD-side windows; conditional healthy-switch blanking is not proven isolation.

J1



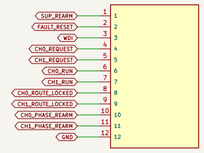
EXTERNAL SUPERVISOR AND HOST RAILS

J2



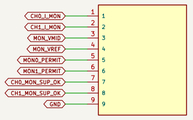
FLOATING AFE RAILS AND OBSERVED CONTROLS

J3



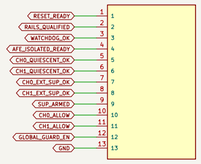
RAW CONTROL / STATUS

J4



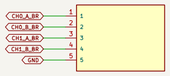
EXISTING MONITOR OBSERVATIONS

J5



EXPORTED NATIVE HEALTH / PERMISSION

J6



DRIVER INTERNAL DIFFERENTIAL VOLTAGES

GLOBAL_GUARD_EN

SUP_3V3

SUP_5

HOST_3V3

HOST_5

HOST_15

VH_24

GND

AFE_3V3

AFE_P5

AFE_N5

AFEAGND

guards
File: guards.kicad_sch

host_rails
File: host_rails.kicad_sch

host_rails_2
File: host_rails_2.kicad_sch

afe_rails
File: afe_rails.kicad_sch

afe_rails_2
File: afe_rails_2.kicad_sch

watchdog
File: watchdog.kicad_sch

quiescence
File: quiescence.kicad_sch

voltage
File: voltage.kicad_sch

duration
File: duration.kicad_sch

admission
File: admission.kicad_sch

admission_2
File: admission_2.kicad_sch

phase0
File: phase0.kicad_sch

phase1
File: phase1.kicad_sch

afe_bank0
File: afe_bank0.kicad_sch

afe_bank1
File: afe_bank1.kicad_sch

afe_bank2
File: afe_bank2.kicad_sch

afe_bank3
File: afe_bank3.kicad_sch

afe_bank4
File: afe_bank4.kicad_sch

afe_logic
File: afe_logic.kicad_sch

afe_logic_2
File: afe_logic_2.kicad_sch

defaults
File: defaults.kicad_sch

Native supervision; converters, firmware, PCB and implant qualification remain

Project ATO

Sheet: /

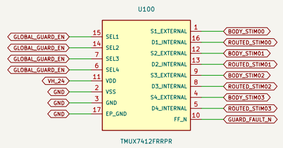
File: supervision.kicad_sch

Title: ATO S4 / Interfaces and handoff

Size: A2	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 1/22

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / NATIVE INDEPENDENT SUPERVISION

18 guarded stimulation conductors, 17 measured RHD-side windows; conditional healthy-switch blanking is not proven isolation.







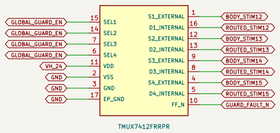






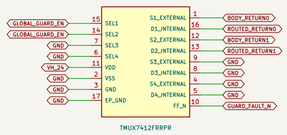


















All net labels are native electrical connections. Actual physical delay, leakage, failed switches and shared supplies remain bounded limitations.

Independent phase cap, heartbeat watchdog, rail windows and deliberate rearm have different functions. No automatic retry or reverse pulse.

Native supervision; converters, firmware, PCB and implant qualification remain

Project ATO

Sheet: /guards/

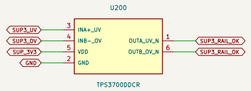
File: guards.kicad_sch

Title: ATO S4 / 18 body-side terminal guards

Size: A2	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 2/22

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / NATIVE INDEPENDENT SUPERVISION

18 guarded stimulation conductors, 17 measured RHD-side windows; conditional healthy-switch blanking is not proven isolation.





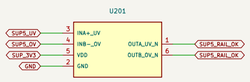














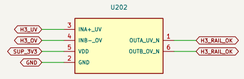
























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Native supervision; converters, firmware, PCB and implant qualification remain

Project ATO

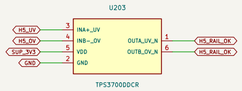
Sheet: /host_rails/
File: host_rails.kicad_sch

Title: ATO S4 / Host supply windows

Size: A2	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 3/22

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / NATIVE INDEPENDENT SUPERVISION

18 guarded stimulation conductors, 17 measured RHD-side windows; conditional healthy-switch blanking is not proven isolation.





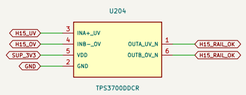














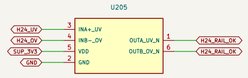
























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Native supervision; converters, firmware, PCB and implant qualification remain

Project ATO

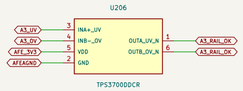
Sheet: /host_rails_2/
File: host_rails_2.kicad_sch

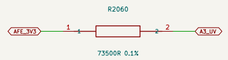
Title: ATO S4 / Host supply windows / 2

Size: A2	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 4/22

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / NATIVE INDEPENDENT SUPERVISION

18 guarded stimulation conductors, 17 measured RHD-side windows; conditional healthy-switch blanking is not proven isolation.



















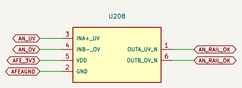














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Native supervision; converters, firmware, PCB and implant qualification remain

Project ATO

Sheet: /afe_rails/
File: afe_rails.kicad_sch

Title: ATO S4 / AFE supply windows

Size: A2	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 5/22

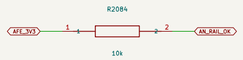
DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / NATIVE INDEPENDENT SUPERVISION

18 guarded stimulation conductors, 17 measured RHD-side windows; conditional healthy-switch blanking is not proven isolation.































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Native supervision; converters, firmware, PCB and implant qualification remain

Project ATO

Sheet: /afe_rails_2/
File: afe_rails_2.kicad_sch

Title: ATO S4 / AFE supply windows / 2

Size: A2	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 6/22

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / NATIVE INDEPENDENT SUPERVISION

18 guarded stimulation conductors, 17 measured RHD-side windows; conditional healthy-switch blanking is not proven isolation.



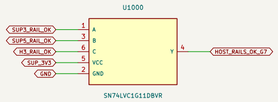




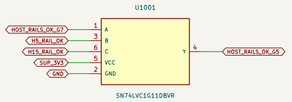


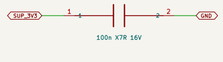


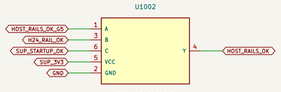




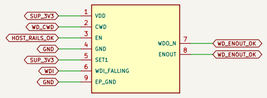


















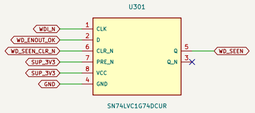




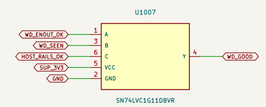


















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Project ATO

Sheet: /watchdog/
File: watchdog.kicad_sch

Title: ATO S4 / Heartbeat and startup

Size: A2	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 7/22

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / NATIVE INDEPENDENT SUPERVISION

18 guarded stimulation conductors, 17 measured RHD-side windows; conditional healthy-switch blanking is not proven isolation.

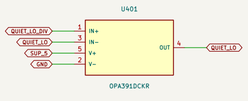




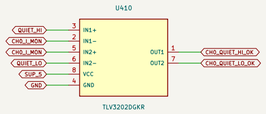




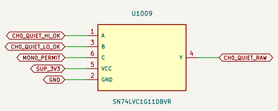






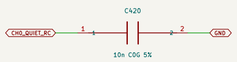










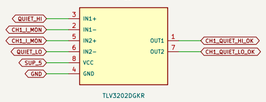


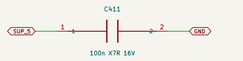


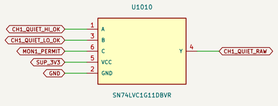












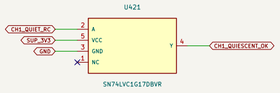














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Project ATO

Sheet: /quiescence/
File: quiescence.kicad_sch

Title: ATO S4 / Current quiet window

Size: A2	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 8/22

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / NATIVE INDEPENDENT SUPERVISION

18 guarded stimulation conductors, 17 measured RHD-side windows; conditional healthy-switch blanking is not proven isolation.

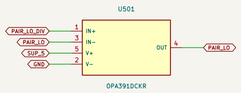




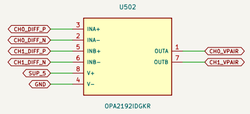




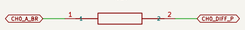






















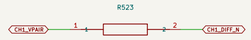


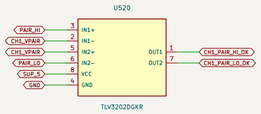




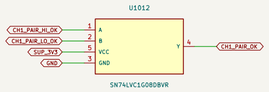














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Native supervision; converters, firmware, PCB and implant qualification remain

Project ATO

Sheet: /voltage/
File: voltage.kicad_sch

Title: ATO S4 / Differential compliance

Size: A2	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 9/22

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / NATIVE INDEPENDENT SUPERVISION

18 guarded stimulation conductors, 17 measured RHD-side windows; conditional healthy-switch blanking is not proven isolation.

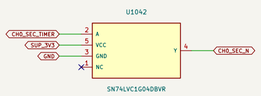




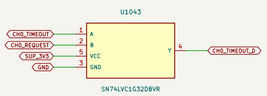
















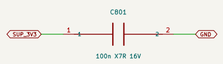






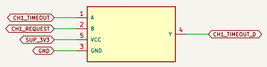




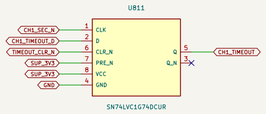














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Native supervision; converters, firmware, PCB and implant qualification remain

Project ATO

Sheet: /duration/

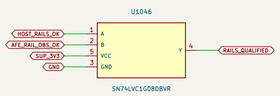
File: duration.kicad_sch

Title: ATO S4 / Secondary phase duration

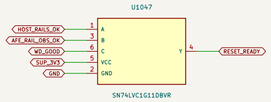
Size: A2	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 10/22

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / NATIVE INDEPENDENT SUPERVISION

18 guarded stimulation conductors, 17 measured RHD-side windows; conditional healthy-switch blanking is not proven isolation.



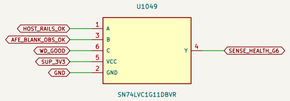




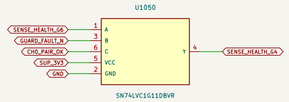
















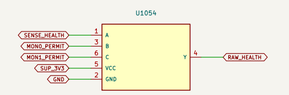




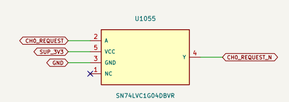


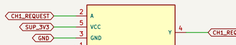




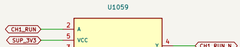














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Native supervision; converters, firmware, PCB and implant qualification remain

Project ATO

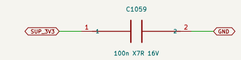
Sheet: /admission/
File: admission.kicad_sch

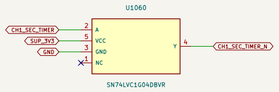
Title: ATO S4 / Sticky permission and rearm

Size: A2	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 11/22

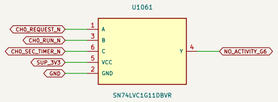
DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / NATIVE INDEPENDENT SUPERVISION

18 guarded stimulation conductors, 17 measured RHD-side windows; conditional healthy-switch blanking is not proven isolation.





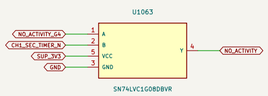




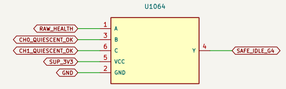




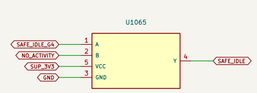




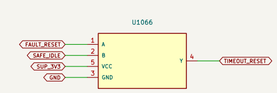
















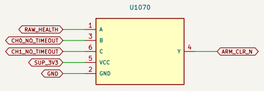








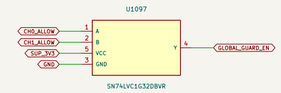














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Native supervision; converters, firmware, PCB and implant qualification remain

Project ATO

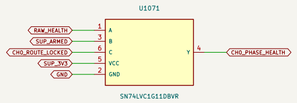
Sheet: /admission_2/
File: admission_2.kicad_sch

Title: ATO S4 / Sticky permission and rearm / 2

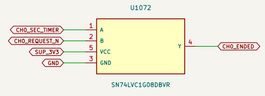
Size: A2	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 12/22

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / NATIVE INDEPENDENT SUPERVISION

18 guarded stimulation conductors, 17 measured RHD-side windows; conditional healthy-switch blanking is not proven isolation.



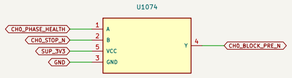








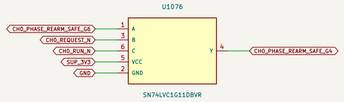












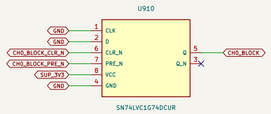








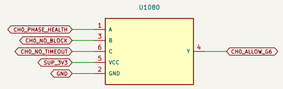








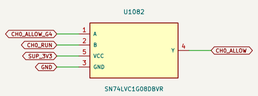




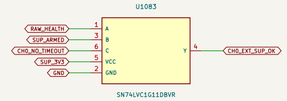














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Project ATO

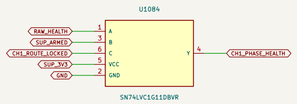
Sheet: /phase0/
File: phase0.kicad_sch

Title: ATO S4 / Source 0 phase permission

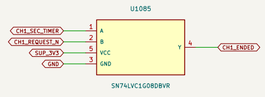
Size: A2	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 13/22

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / NATIVE INDEPENDENT SUPERVISION

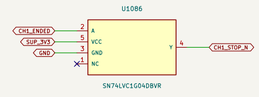
18 guarded stimulation conductors, 17 measured RHD-side windows; conditional healthy-switch blanking is not proven isolation.



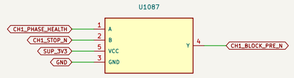




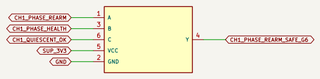
















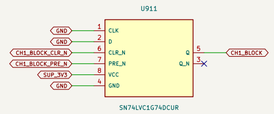








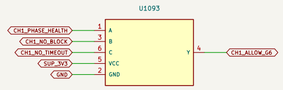




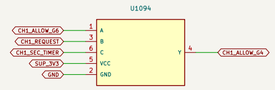




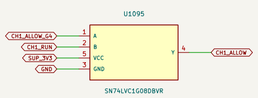




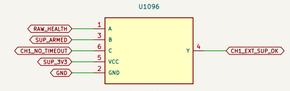














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Project ATO

Sheet: /phase1/
File: phase1.kicad_sch

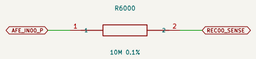
Title: ATO S4 / Source 1 phase permission

Size: A2	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 14/22

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / NATIVE INDEPENDENT SUPERVISION

18 guarded stimulation conductors, 17 measured RHD-side windows; conditional healthy-switch blanking is not proven isolation.

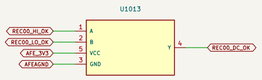




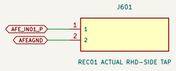




















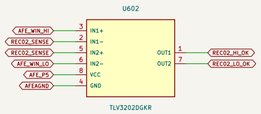




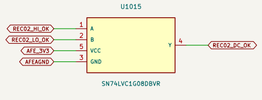




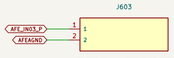






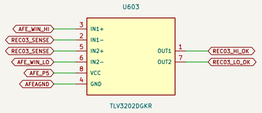


















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Project ATO

Sheet: /afe_bank0/
File: afe_bank0.kicad_sch

Title: ATO S4 / RHD input windows 0

Size: A2	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 15/22

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / NATIVE INDEPENDENT SUPERVISION

18 guarded stimulation conductors, 17 measured RHD-side windows; conditional healthy-switch blanking is not proven isolation.

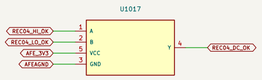




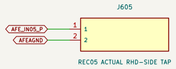






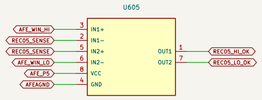








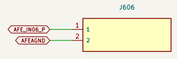






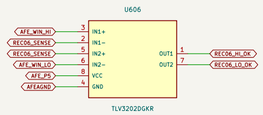
































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Native supervision; converters, firmware, PCB and implant qualification remain

Project ATO

Sheet: /afe_bank1/
File: afe_bank1.kicad_sch

Title: ATO S4 / RHD input windows 1

Size: A2	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 16/22

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / NATIVE INDEPENDENT SUPERVISION

18 guarded stimulation conductors, 17 measured RHD-side windows; conditional healthy-switch blanking is not proven isolation.

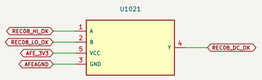




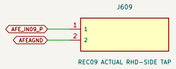






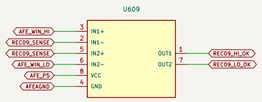








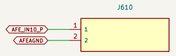






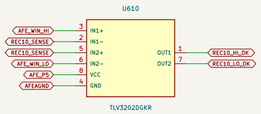




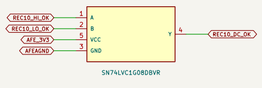




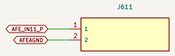






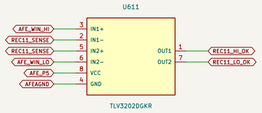


















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Native supervision; converters, firmware, PCB and implant qualification remain

Project ATO

Sheet: /afe_bank2/
File: afe_bank2.kicad_sch

Title: ATO S4 / RHD input windows 2

Size: A2	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 17/22

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / NATIVE INDEPENDENT SUPERVISION

18 guarded stimulation conductors, 17 measured RHD-side windows; conditional healthy-switch blanking is not proven isolation.

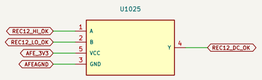




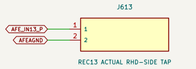






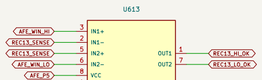








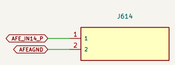






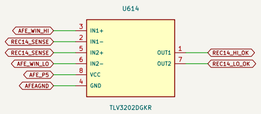






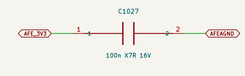


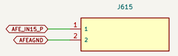








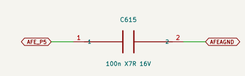


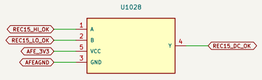














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Native supervision; converters, firmware, PCB and implant qualification remain

Project ATO

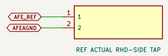
Sheet: /afe_bank3/
File: afe_bank3.kicad_sch

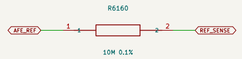
Title: ATO S4 / RHD input windows 3

Size: A2	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 18/22

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / NATIVE INDEPENDENT SUPERVISION

18 guarded stimulation conductors, 17 measured RHD-side windows; conditional healthy-switch blanking is not proven isolation.















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Project ATO

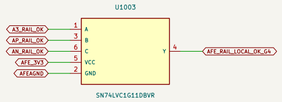
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File: afe_bank4.kicad_sch

Title: ATO S4 / RHD input windows 4

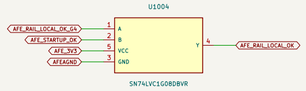
Size: A2	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 19/22

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / NATIVE INDEPENDENT SUPERVISION

18 guarded stimulation conductors, 17 measured RHD-side windows; conditional healthy-switch blanking is not proven isolation.











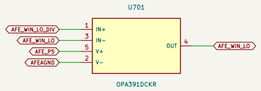






























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Project ATO

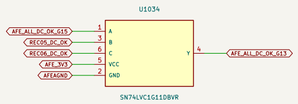
Sheet: /afe_logic/
File: afe_logic.kicad_sch

Title: ATO S4 / AFE blank and isolated status

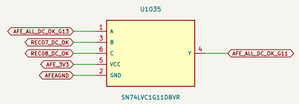
Size: A2	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 20/22

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / NATIVE INDEPENDENT SUPERVISION

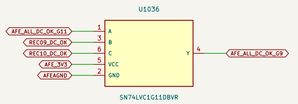
18 guarded stimulation conductors, 17 measured RHD-side windows; conditional healthy-switch blanking is not proven isolation.

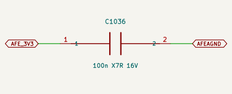


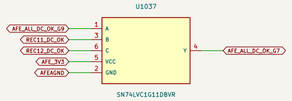


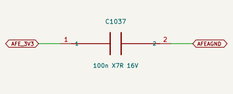


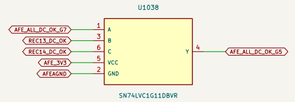


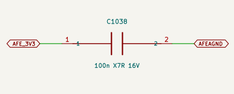


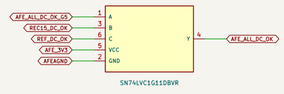








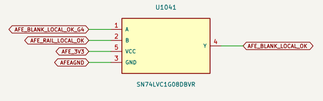




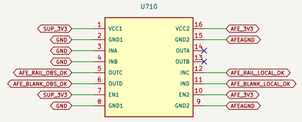
















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Project ATO

Sheet: /afe_logic_2/
File: afe_logic_2.kicad_sch

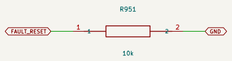
Title: ATO S4 / AFE blank and isolated status / 2

Size: A2	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 21/22

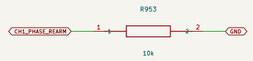
DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / NATIVE INDEPENDENT SUPERVISION

18 guarded stimulation conductors, 17 measured RHD-side windows; conditional healthy-switch blanking is not proven isolation.



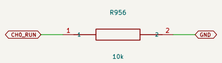






















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Project ATO

Sheet: /defaults/
File: defaults.kicad_sch

Title: ATO S4 / External defaults

Size: A2	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 22/22